NHS ENGLAND

General Practice Appointment Analysis

Diagnostic Analysis of NHS Appointment Efficiency

Using Python — Technical Validation Report

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COURSE LSE Business Analytics for Executives | DA201

PERIOD NHS GP Appointments — August 2021 to June 2022

TOOLS Python · Pandas · Seaborn · Matplotlib · NumPy

4	0	84.5%	74.8%	30.4M
Datasets Analysed	Missing Values	Peak Utilisation	Avg. Utilisation	Peak Month Appointments

1. Introduction & Background

The National Health Service (NHS) faces a persistent operational challenge from missed GP appointments — formally classified as DNA (Did Not Attend) events. This report documents the analytical approach, Python-based methodology, and validated findings derived from four NHS-published datasets, providing technical validation of results to inform evidence-based decision-making. Two core mandates framed this investigation: **(1)** whether adequate staff and capacity existed within NHS networks, and **(2)** the actual utilisation of available resources. The analysis was conducted in a Jupyter Notebook using Pandas, Seaborn, Matplotlib, and NumPy.

1.1 Dataset Overview

Alias	File	Key Variables	Rows	Date Range
AD	actual_duration.csv	Location, date, duration, count	137,793	Dec 2021–Jun 2022
AR	appointments_regional.csv	Status, HCP type, mode, booking lead time	596,821	Jan 2020–Jun 2022
NC	national_categories.xlsx	Service setting, context type, nat. category	817,394	Aug 2021–Jun 2022
TW	tweets.csv	Tweet text, hashtags, retweet/fav counts	1,174	N/A

2. Analytical Approach & Data Quality

Each dataset was imported as a Pandas DataFrame and sense-checked using `df.shape()`, `df.dtypes()`, `df.isnull().sum()`, `df.describe()`, and `df.duplicated()`. A critical finding was that **all four datasets contained zero missing values and zero duplicate records**, confirming data integrity. Date fields were parsed to `datetime64` format using `pd.to_datetime()`. Structural exploration using `pd.unique()` confirmed **106 sub-ICB locations**, 42 ICBs across 7 NHS regions. The NC dataset contained **5 service settings**, **3 context types**, **18 national categories**, and **3 appointment statuses**. The analysis window was aligned to **August 2021 to June 2022**. An important caveat from the NHS metadata: GP systems were not designed for analytical reporting, and data entry varies across practices. The **1,200,000 appointments per day** capacity benchmark was used as the upper threshold for all utilisation calculations.

3. Findings: Capacity & Staffing (Mandate 1)

3.1 Monthly Appointment Volume

Using `groupby('appointment_month')` on the AR dataset (filtered to August 2021 onwards), monthly appointment totals were computed. Figure 1 plots these against the 36M monthly capacity ceiling. A clear **bimodal seasonal pattern** emerges: an autumn peak in October–November 2021 and a secondary spring peak in March 2022. **November 2021 was the single busiest month at 30,405,070 appointments**, followed by October 2021 at 30,303,834. Critically, **the 36M monthly capacity threshold was never breached** during the analysis period.

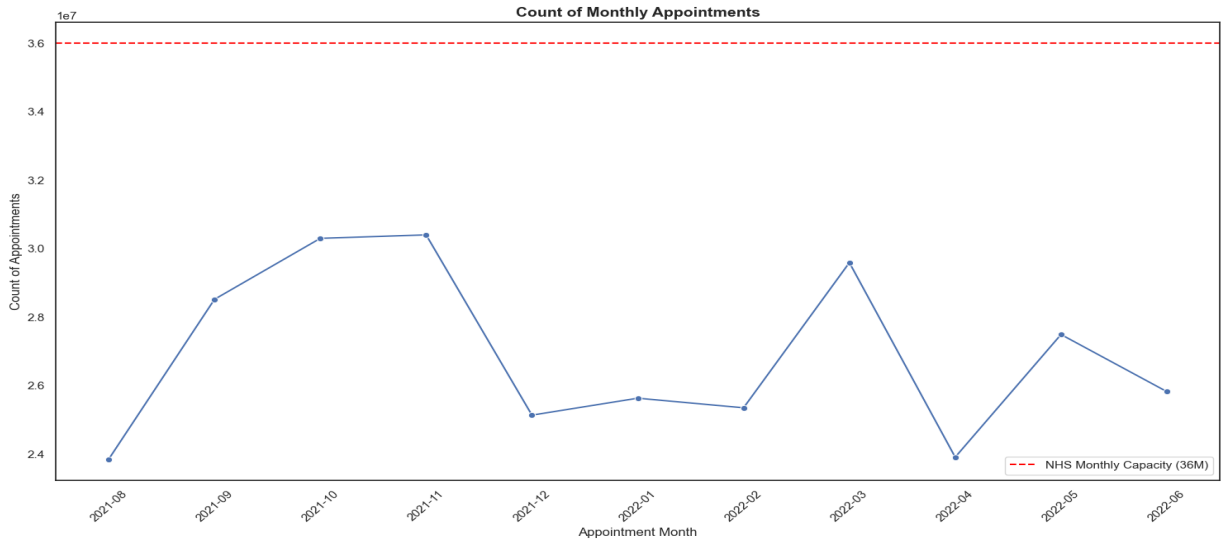


Figure 1: Count of Monthly Appointments vs. NHS Capacity (36M/month)

3.2 Capacity Utilisation

Utilisation was computed as: monthly total ÷ 30 days ÷ 1,200,000 × 100. Figure 2 confirms **the 1,200,000/day threshold was never exceeded**. **Peak utilisation reached 84.5% in November 2021** (daily average: 1,013,502 appointments), with a period average of **74.8%** — maintaining approximately 25% headroom below maximum capacity. This directly addresses Mandate 1: a blanket national staffing increase is not supported by the aggregate data. However, regional sub-ICB analysis may reveal localised hotspots, particularly during October–November.

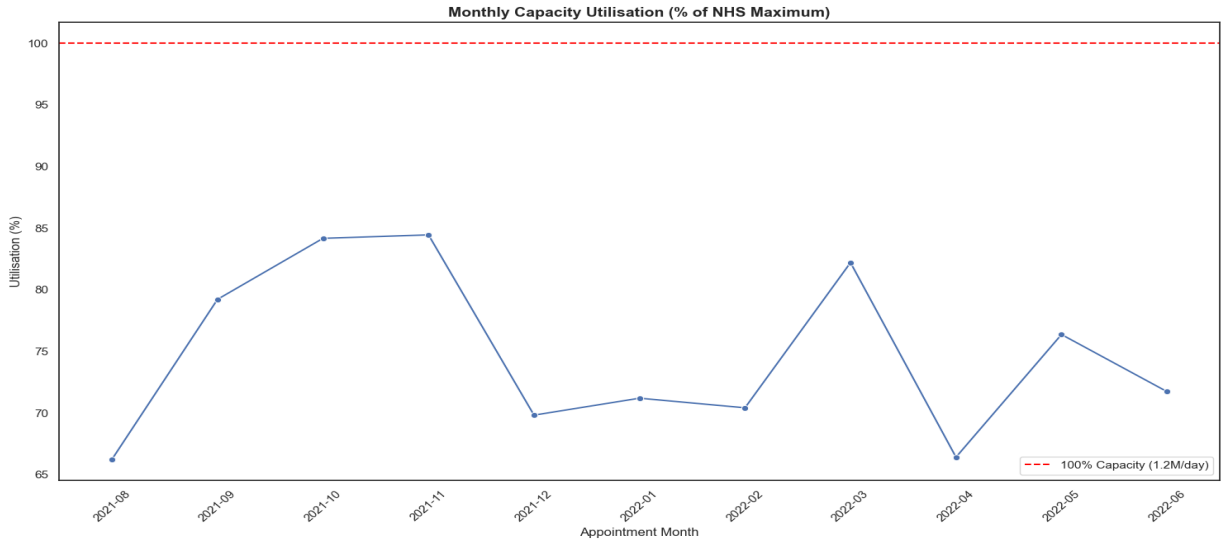


Figure 2: Monthly Capacity Utilisation (% of 1.2M/day maximum)

4. Findings: Resource Utilisation (Mandate 2)

4.1 Appointment Attendance & DNA Events

Figures 3 and 4 show appointment status trends over time. The attendance rate was **91.4%** across the period. DNA appointments spiked significantly in **October 2021 (1,565,624)** and **November 2021 (1,428,087)** — exactly the months with the highest overall volumes. This positive correlation is analytically important: the NHS busiest periods are simultaneously when it loses most capacity to no-shows. The predictability of these spikes creates a clear opportunity for proactive intervention.

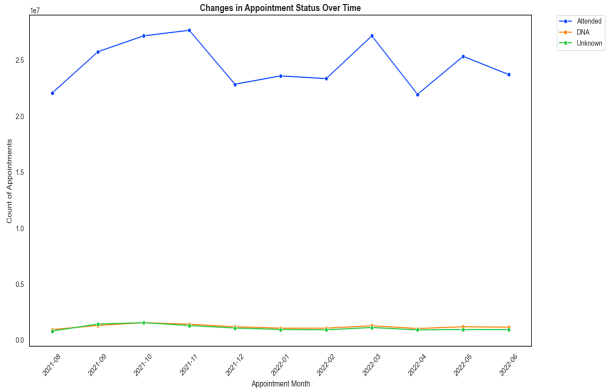


Figure 3: Appointment Status Over Time

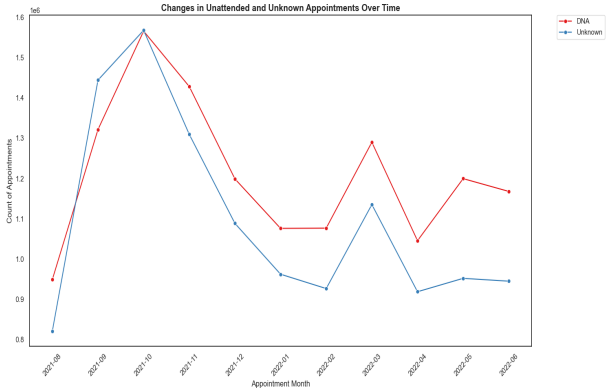


Figure 4: DNA & Unknown Appointments Over Time

4.2 HCP Types Over Time

Figure 6 shows GP and Other Practice Staff following near-identical seasonal patterns, both peaking in October/November 2021 and March 2022 before declining. **This parallel movement confirms that demand — not staffing changes — drives variation.** There is no evidence of a structural shift in HCP type distribution during the period, reinforcing that staffing levels were broadly proportionate to observed demand.

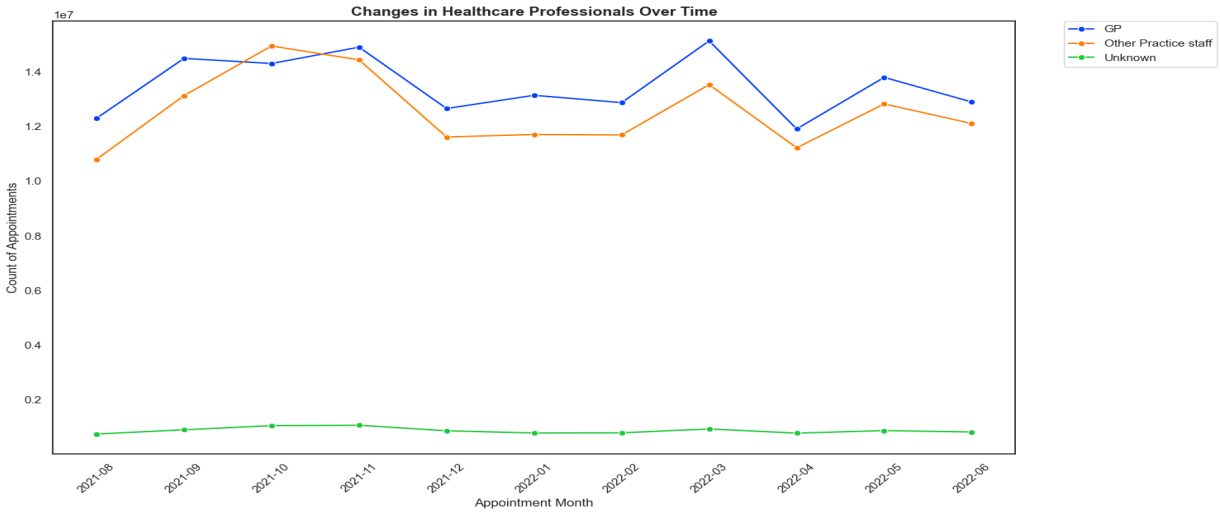


Figure 6: Changes in Healthcare Professionals (HCP Types) Over Time

4.3 Appointment Mode

Figure 7 confirms that **Face-to-Face appointments are the dominant mode at 62.0%**, with Telephone at 33.9%. During the busiest months, face-to-face demand increases while telephone declines slightly, suggesting patients prefer in-person care during demand surges. **Home Visit and Video/Online appointments remain consistently low throughout**, representing an underutilised efficiency opportunity.

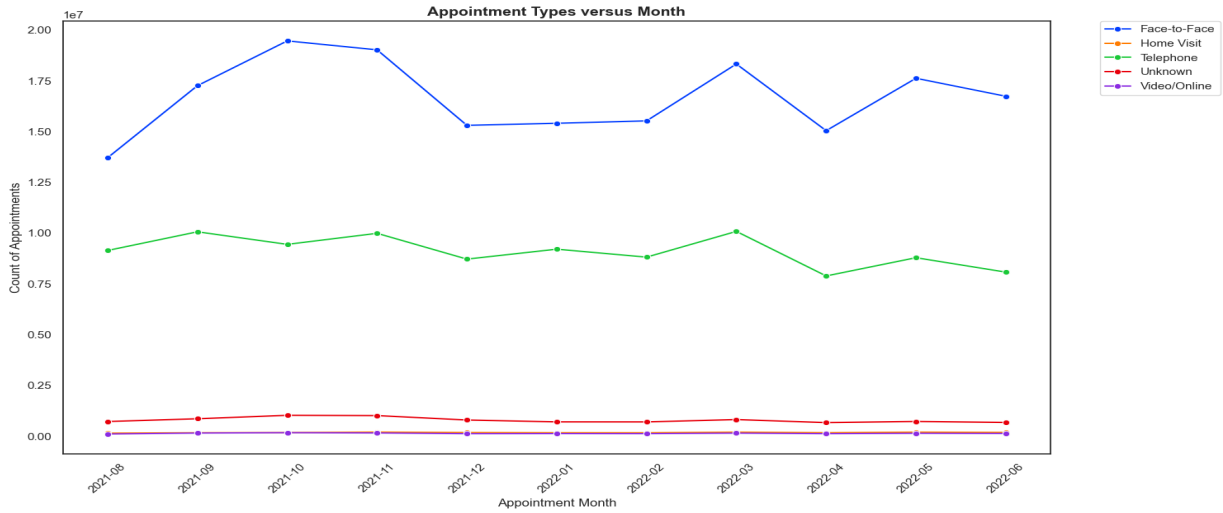


Figure 7: Appointment Types (Mode) by Month

4.4 Time Between Booking and Appointment

Figure 8 shows that **Same Day bookings are the most common category at 44.2%**, confirming the predominantly reactive nature of NHS primary care. The 2–7 day category follows at 20.5%. Appointments booked **8–28 days in advance increased during October 2021 and March 2022**, potentially indicating same-day slot exhaustion — a leading indicator of localised capacity strain warranting ongoing monitoring.

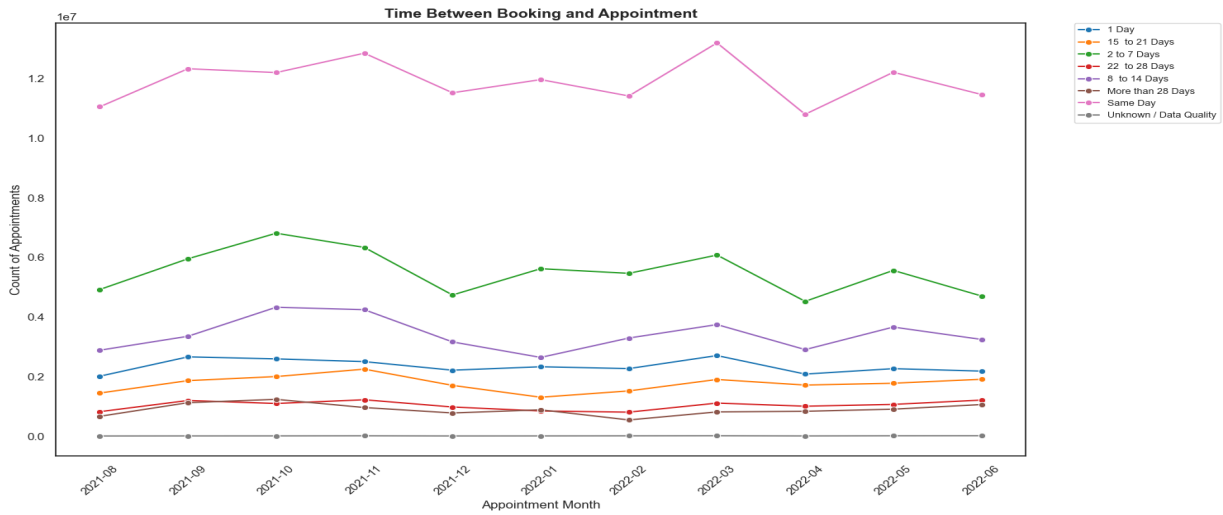


Figure 8: Time Between Booking and Appointment

4.5 Service Settings Comparison

Figures 9 and 10 present boxplots of daily appointment counts by service setting. Figure 9 confirms **General Practice dominates at ~91.5% of all appointments**. Figure 10, excluding GP, reveals that **Unmapped** shows the widest spread with a declining trend — reflecting improving NHS data recording practices — while **Primary Care Network (PCN)** appointments grow, evidencing measurable impact from the Additional Roles Reimbursement Scheme (ARRS).

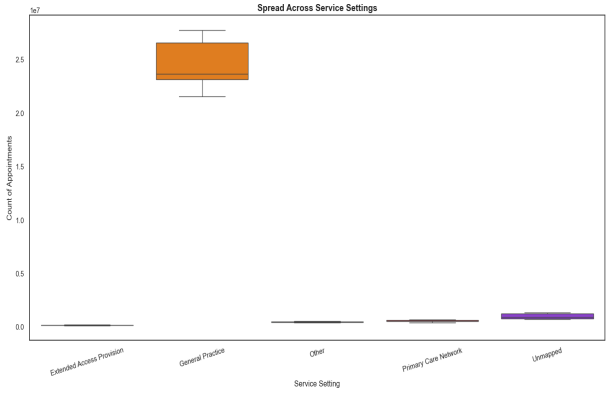


Figure 9: Spread Across All Service Settings

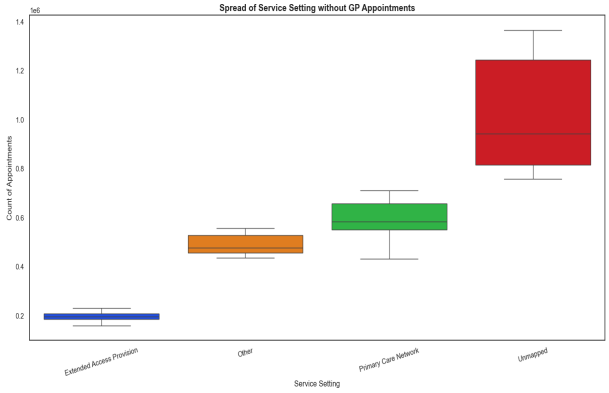


Figure 10: Spread of Service Settings (Excl. General Practice)

5. Twitter/X Social Media Analysis

The 1,174-row Tweets dataset was analysed by looping through `tweet_full_text` to extract hashtags, converting to frequency counts using `pd.Series().value_counts()`, and filtering to count > 10 (47 hashtags). Figure 5 shows the distribution excluding **#healthcare**, which dominated at **769 occurrences** — more than eight times the next tag. Key clusters: **#health (83)** and **#medicine (42)** confirm broad public engagement; **#ai (44)** and **#digitalhealth (29)** signal growing NHS technology discourse; **#job (38)** and **#hiring (18)** corroborate staffing concerns. Twitter/X represents a direct, cost-effective channel for targeted campaigns ahead of predicted DNA spike periods.

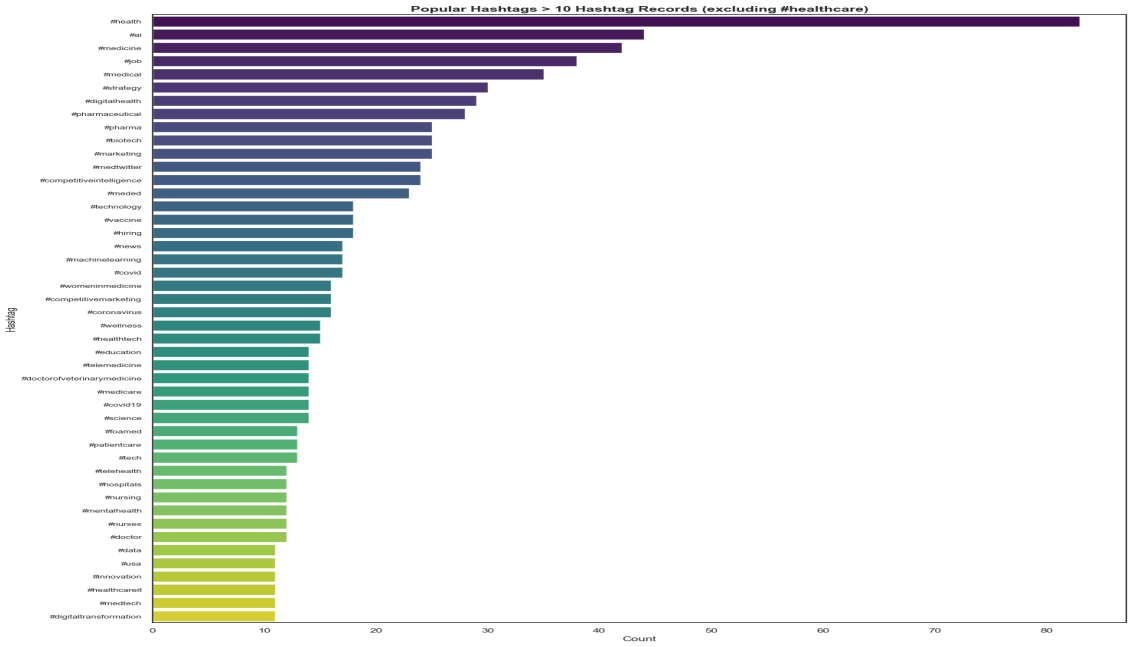


Figure 5: Popular Healthcare Hashtags (count > 10, excluding #healthcare — 769 occurrences)

6. Conclusions & Recommendations

This analysis provides data-validated answers to both NHS mandates. National capacity (1.2M/day) was **never exceeded** during August 2021–June 2022, with peak utilisation at 84.5% and a period average of 74.8%. A blanket increase in NHS staffing is not supported by the aggregate data; the evidence instead points to targeted, efficiency-driven interventions. Seven recommendations follow:

1. Do not broadly increase staffing: Utilisation averages 74.8% nationally — investigate sub-ICB and seasonal hotspots, particularly October–November and March, before committing to staffing increases.

2. Proactively reduce DNA appointments: DNA spikes in October 2021 and March 2022 are predictable and repeat annually. Deploy SMS, email, and Twitter/X campaigns 2–4 weeks ahead of these peaks, focused on patient impact rather than financial penalties.

3. Pilot extended weekend access: Near-zero weekend appointments create Monday bottlenecks. Weekend provision pilots in high-volume sub-ICBs could smooth weekly demand distribution.

4. Promote telephone and video appointments: Video/Online modes are consistently underutilised. Active promotion during peak periods reduces face-to-face congestion and infection risk at no additional staffing cost.

5. Monitor 8–28 day booking lead times: Rising advance bookings during peak months signal same-day slot exhaustion. Track as a leading capacity strain indicator before it becomes a crisis.

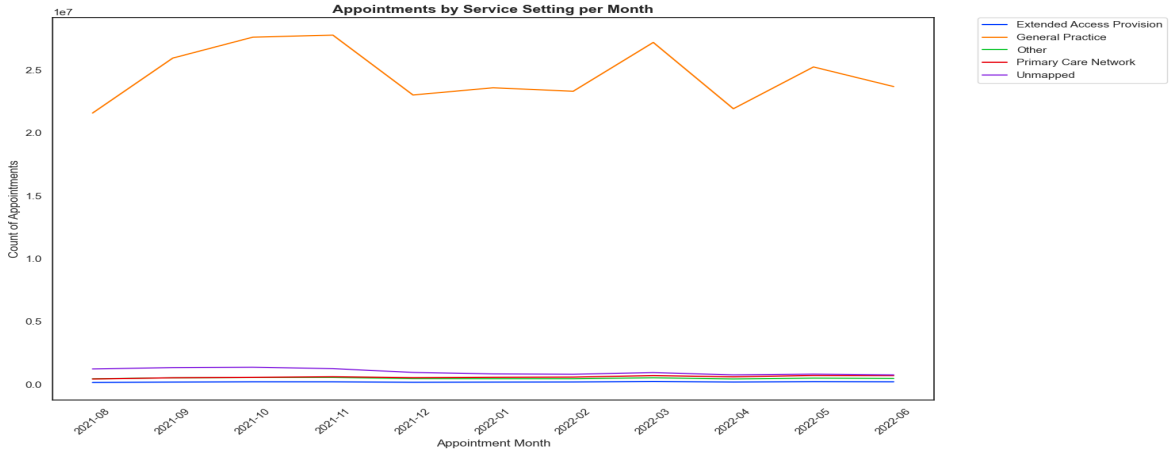
6. Use Twitter/X strategically: The confirmed public engagement with #healthcare (769 occurrences) provides a direct channel to amplify NHS communications ahead of predicted DNA spikes.

7. Accelerate NHS data standardisation: Declining Unmapped appointments show progress. Continuing NHS-wide standardisation of GP system recording will improve the accuracy of future analytical decision-making.

Appendix: Supporting Visualisations

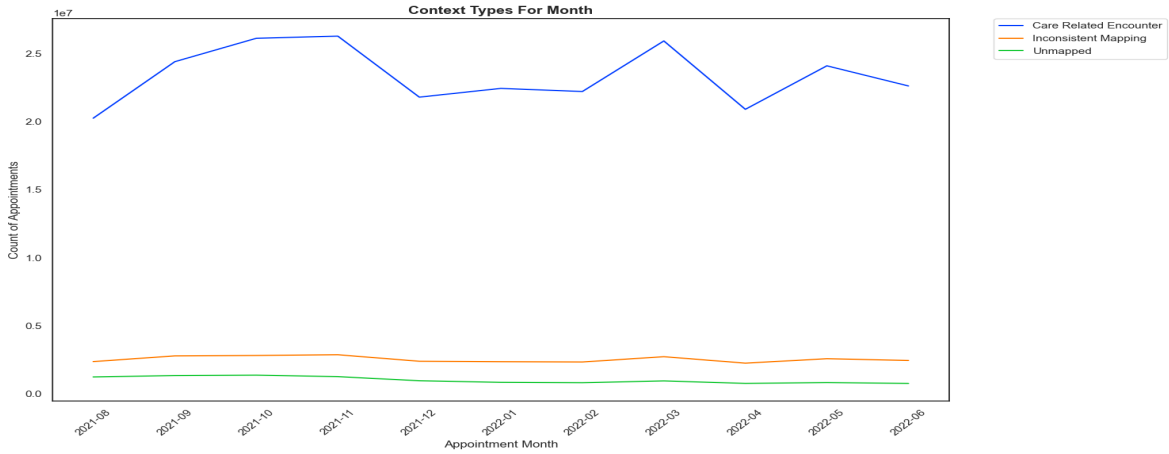
The following charts were produced in Activity 4 (monthly and seasonal trends) and Activity 5 (full hashtag distribution) of the Jupyter Notebook. They support the analytical narrative in the main body and are included here for completeness.

Figure A1: Appointments by Service Setting per Month (Activity 4, Objective 1)



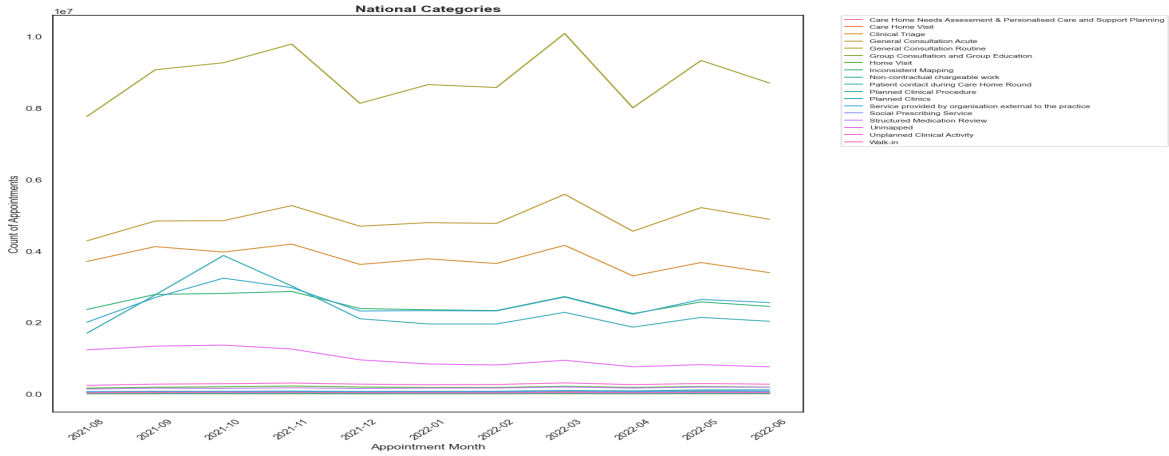
Observation: General Practice dominates (~91.5%). Unmapped shows a declining trend. PCN appointments rise across the period.

Figure A2: Context Types per Month (Activity 4, Objective 1)



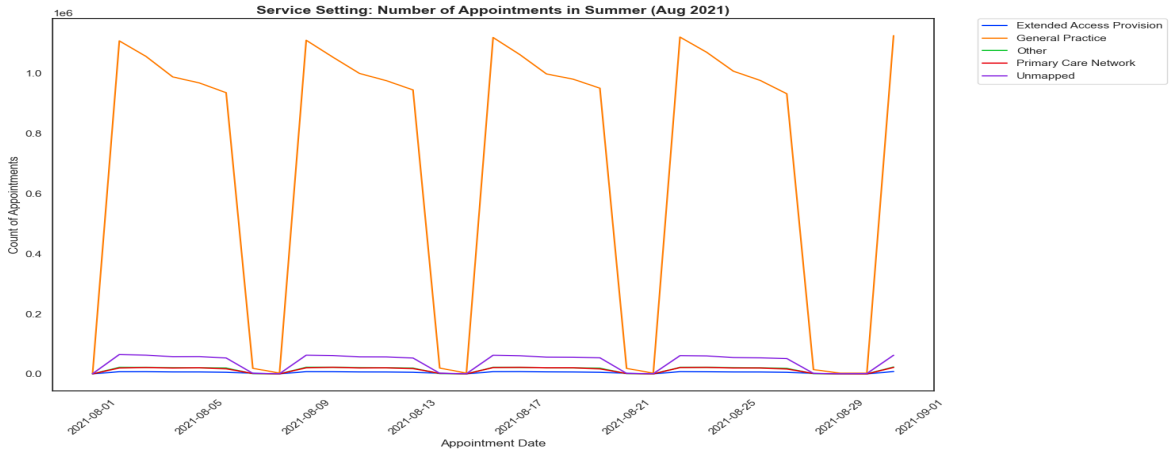
Observation: Care Related Encounter accounts for ~86.8% of all appointments, following the same bimodal seasonal pattern.

Figure A3: National Categories per Month (Activity 4, Objective 1)



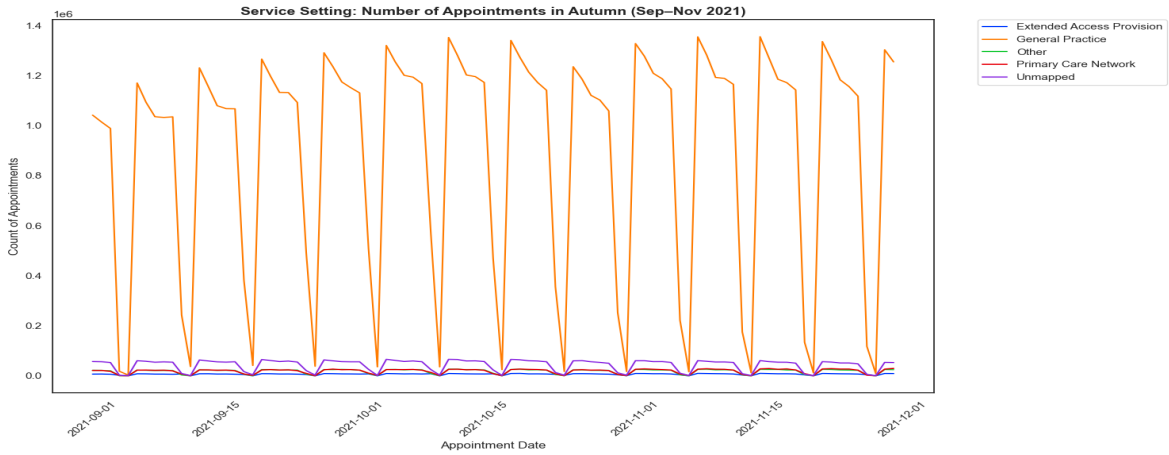
Observation: General Consultation Routine is the most common category (~32.9%), followed by General Consultation Acute (~18.1%). 18 categories total.

Figure A4: Service Setting: Summer 2021 (August only) — Activity 4, Objective 2



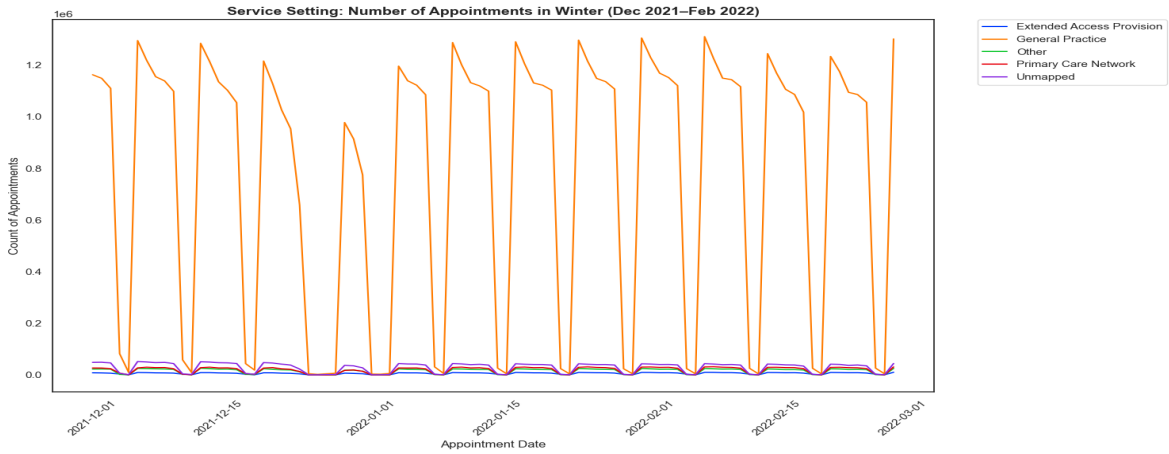
Observation: The weekly sawtooth pattern is clearly visible: Monday peaks, near-zero at weekends. Summer records the lowest appointment volumes.

Figure A5: Service Setting: Autumn 2021 (Sep–Nov) — Activity 4, Objective 2



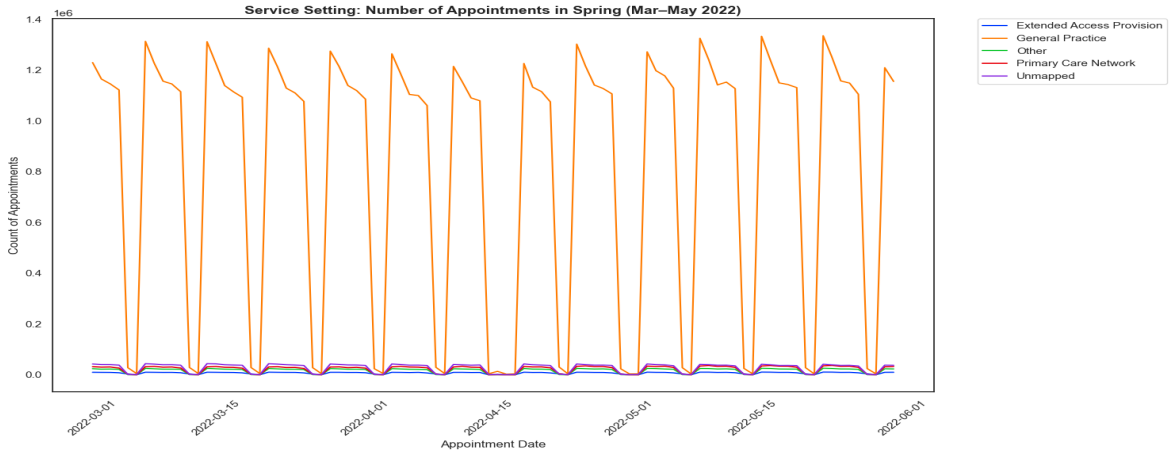
Observation: Busiest season. GP appointments regularly approach the daily capacity limit. October and November record the highest counts.

Figure A6: Service Setting: Winter 2021/22 (Dec–Feb) — Activity 4, Objective 2



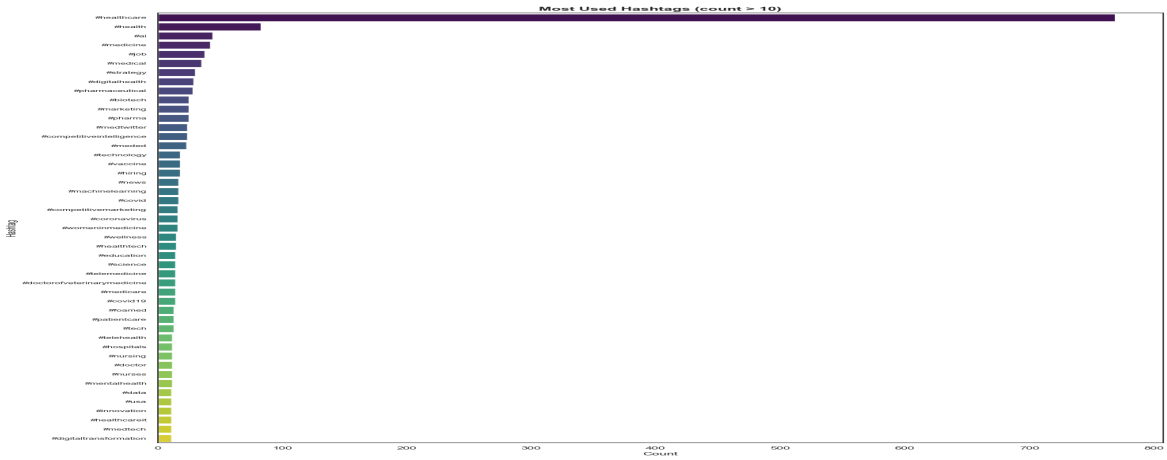
Observation: A clear Christmas/New Year dip is visible in late December, followed by a strong January recovery.

Figure A7: Service Setting: Spring 2022 (Mar–May) — Activity 4, Objective 2



Observation: March 2022 is the second-highest peak, coinciding with full lifting of COVID-19 restrictions. An Easter dip is visible in April.

Figure A8: All Healthcare Hashtags with Count > 10 (Activity 5)



Observation: #healthcare dominates at 769 occurrences — excluded from Figure 5 in the main report to reveal the full distribution below it.